

REPORT ON JHEP_027P_0726

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Title: Modeling Falling Backgrounds with Exponential Mixtures

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Report of referee 1

Abstract "motivate the finite exponential mixture as a" This phrase is not self-explanatory. I was imagining some sort of $e * a + bx + cx^2$ Googling finds your article as top hit, so it does not seem to be in common use.

I misinterpreted 'n= 28, 619, 185 and n = 5, 036' as somehow identifying numbers of a dataset, not the size.

Section 2 starts with pure motivations, but quickly reverts into an ad-hoc model. But having explained the benefits of the GPD in including models with upper cut-offs, these are not supported.

Eqn 2.4 is rather straightforward and its not clear what is gained by passing through the continuous eqn 2.2

"In some regimes the Normal distribution tail converges to the GPD" I am not clear what we learn from this - you moved away from the GPD, right?

At the top of p5 you make it clear that the convergence is not straight-forward 'multiple random restarts'. This sounds slow and cumbersome. You need a procedure to define success...what is it?

You introduce AIC/BIC. It is very common in the field to use Wilkes tests, or the derivative $2 * \Delta(\text{ll}) < X$ when testing different numbers of parameters. Did you have a good reason to use a different approach?

on p6 you write 80 restarts and 100 retries 'since the dataset is large'. This does not seem like an automated procedure.

table 1/2: why are you not reporting the Ndf of the fit and order of the Exponential mixture? Its not clear why you quote the chi2 here at all - the ll does not need this arbitrary rebinning to have 30 events and is just as informative for comparing values. Indeed for the diphoton dataset

the visually obvious deviations are above 2000 and are perhaps lost in the rebinning.

page 10 "Since fits to exponential mixtures of varying complexity may have weakly identified parameters," What do you mean? Anticorrelated parameters may give unstable results? This is a worrying flag that it would be good to explain a bit more. The impact is that you try to estimate coverage of the simulation using bootstrap on the data. This is a fine approach but does but measures not true density but fluctuations of the observed data, which is not quite the same thing - if your dataset is 'up' in one place the sampling will vary around that 'up'. But there is no foolproof solution here.

Metrics: you discuss cooverage, spurious signal and Bias, all of which are interesting, but you do not compare the power of the different approaches, which is an unfortunate omission. AIC and BIC for example will surely differ in power.

The Discussion is misleading. You had to abandon the basis functions drawn from EVT, so you should not now appeal to it to support your proposal.

The rest of the discussion i applaud, bar the lack of discussion of sensitivity - the price paid for multiple parameters. Without that there is really no reason not to go to $N \rightarrow \text{infinity}$. Without this you miss an important test.

Fig 2 and 4. Perhaps I have misunderstood something in your procedure? As we take the limit of large x the exponential with the smallest weight β will increase in fraction. The w parameters are by construction non-negative. So surely the log of the function has to be concave, which is not the case in these figures. ? Is this not implicit in your monotone discussion? If negative W 's are allowed there is of course more freedom..but also more fine tuning to avoid negative PDFs.

Report of referee 2

This contribution introduces and motivates finite exponential mixtures as an approach for modeling smoothly falling spectra, for applications such as resonance searches at the LHC. It summarizes the relevant background from extreme value theory before applying the idea to two examples: an ATLAS dijet spectrum and a CMS diphoton case. The performance of the approach is compared against the approaches used by the respective collaborations. Simulation-based studies then provide a more detailed evaluation in the CMS setup, highlighting low bias, relatively good coverage, and controlled levels of spurious signal.

The work presented is a useful, even if incremental, contribution to the field: with increasing dataset sizes, functional-form-based background mod-

eling is becoming increasingly difficult. The paper adds a new promising direction on top of the cited existing approaches, summarizes the background concisely, and motivates the use of finite exponential mixtures both from fundamental statistics considerations and using the studies performed. The individual ingredients are standard, but the paper brings them together and applies them to a concrete HEP challenge.

The main limitations of the paper and where I would have liked to see more are:

- an extended discussion of the under-coverage in tail regions and potential remedies, given that modeling tails is one motivation of the approach;
- a comparison to other approaches, especially the GP-based one in ref [9];
- experiments with signals injected;
- experiments with larger n , especially given the motivation of increasing dataset sizes.

At a technical level, the approach is sound, well motivated, and described well. The paper also appropriately discusses limitations of using finite exponential mixtures. The text and figures are clear. I list a few minor text-related comments below. I recommend minor revisions. While the additional studies I mention would be nice to see, the content is already substantial.

major comments

- The lack of experiments with larger dataset sizes is unfortunate, as most concern the rather small CMS diphoton dataset case. Did you attempt simulation studies with the ATLAS one as well? A result similar to the integrated luminosity scaling in ref [9] would serve as a great motivation, as it stands this feels like a gap to me.
- The paper describes background modeling and spurious signal studies, but does not report on signal injection tests. Adding flexibility to a background model can result in loss of power for a signal search and corresponding experiments would be helpful. Both likelihood ratio studies and linearity / extracted-vs-true signal tests would strengthen this paper.

minor comments

- title page: arXiv identifier is a placeholder, strange line break before "USA"

- "extreme value theory" is sometimes capitalized and sometimes not
- sec 1: "Higgs mass occurs within the bulk of the background" is awkward phrasing, suggest "resonance" or similar
- sec 2: this is very statistics theory focused and the reader may think that exponentials have not been used at all in practice, would be useful to mention here that exponential sums are already common practice (as the later examples show)
- sec 3.1: superfluous "be" in "expected to be require"
- sec 3.1 (and following): how do you convince yourself that 80 random restarts and 100 retries is a sensible choice and that you get sufficiently close to the desired (global) minimum? The discussion section hints at robust optimization being an open question, do you have any practical advice to incorporate here based on the experiments in this paper?
- sec 3.1: "spanning from 2015 to 2018 ." extra space before period
- table 1: would suggest stating that the ν in χ^2_ν is the number of degrees of freedom (notation might not be obvious to all the HEP audience) and ideally also stating what that number is for completeness
- fig 1: after table 1 I was trying to understand how the dijet function enters here until I realized this is only showing exponential mixtures, may help mentioning in the text introducing the figure explicitly that this is only showing those
- fig 1 and fig 3: would be nice to replace with a vector graphic version if available, though perfectly readable at current resolution
- fig 5: legend placement overlaps a bit with the figure content, looks readable enough to me but perhaps can be tuned slightly
- sec 4: "the functions in eq. (3.4)" should be 3.4 to 3.7 (also again later)
- sec 4: I think a brief explanation of label switching is useful here for a more general audience
- sec 4: is there a re-optimization of k done for each resample and does the effect of that choice therefore also enter in the CIs?
- sec 4: how trustworthy is the Hessian-based estimate of σ_s in the spurious signal section, given that the previous coverage section discusses the Hessians are unstable?

- sec 4: typo "regio;,"
- fig 7: caption is about $S(x)$ but figure is N_s/σ_{N_s}
- sec 5: EVT abbreviation is re-defined here
- sec 5: stray "due;,"
- ref [25] uses a different title formatting compared to the rest of the references
- ref [25] uses "Collaboration", while the rest uses "collaboration"